2.1 Conditional Statements

Name _____ Period ____ Date _____

Geometry 2.1 · Student sheet

I can…

- identify the hypothesis and conclusion of a conditional statement.
- write the converse, inverse, and contrapositive and decide whether each is true.
- write a biconditional statement.

Vocabulary · fill in the missing word

Conditional statement — a logical statement with a _____ and a conclusion, written "if p, then q"

Converse — the statement formed by switching the _____ and the conclusion

Inverse — the statement formed by negating both the _____ and the conclusion

Contrapositive — the statement formed by switching and negating the _____ and the conclusion

Biconditional statement — a statement that _____ the phrase "if and only if"

Part 1 · Discovery: true one way, false the other?

Is each statement true? Switch the "if" and "then" parts. Is it still true?

Statement	True?	Switched	True?
If an animal is a dog, then it has four legs.			
If an angle measures 90°, then it is a right angle.			
If a number is even, then it ends in 4.			

One example that shows a statement is false is called a _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Conditional statement: "if p, then q"; p is the _____ and q is the _____.

Example 1

Identify the hypothesis and the conclusion: "If a polygon has 5 sides, then it is a pentagon."

Example 2

Rewrite in if-then form: "All 90° angles are right angles."

Related conditionals: converse $q \rightarrow p$, inverse $\sim p \rightarrow \sim q$, contrapositive _____. A conditional and its _____ are always both true or both false.

Example 3

Write the converse, inverse, and contrapositive of “If two angles form a linear pair, then they are supplementary.” Decide whether each is true.

Example 4

Is the statement true? If not, give a counterexample: “If $x^2 = 16$, then $x = 4$.”

Biconditional: when a conditional and its converse are both true, combine them with the words _____.

Example 5

Write as a biconditional: “If two lines intersect to form a right angle, then they are perpendicular.” (Its converse is also true.)

Part 3 · Partner practice

Partner 1

Write the converse of “If a figure is a square, then it has four sides.” Is the converse true?

Partner 2

Write the contrapositive of “If $m\angle A = 30°$, then $\angle A$ is acute.” Is it true?